BRAC UNIVERSITY

Inspiring Excellence

Department of Mathematics and Natural Sciences
MAT110 : Differential Equations and Coordinate Geometry

Assignment-2

All Questions Contain 10 mark **Total= $15 \times 10 = 150$**

Partial derivatives and Chain rules for partial derivatives

1. Find f_{xx}, f_{xy}, f_{yy}
   (a) $f(x,y) = e^{x^2+xy+y^2}\sin(x^2y)$.
   (b) $f(x,y) = x^3\ln\sin(xy) + x^5y^2 - e^{3x}y^2$.
2. Find $f_{xx}, f_{xy}, f_{yy}, f_{xz}, f_{zz}, f_{yz}, f_{xyz}, f_{xyy}, f_{zyy}, f_{yxx}$ for $f(x,y,z) = x^3\ln\sin(x^2yz) + \tan(xz) - e^{3x}y^2$
3. Let $w = x^2 + y^2 - z^2$ and $x = \rho\sin\phi\sin\theta,\ y = \rho\sin\phi\cos\theta,\ z = \rho\cos\phi$. Find $w_\phi,\ w_\rho,\ w_\theta$
4. Show that $u(x,t) = sin(x-ct) + \ln(x-ct)$ satisfies $u_{tt} = c^2u_{xx}$.
5. Verify that the function $u(x,y,z) = \frac{1}{\sqrt{x^2+y^2+z^2}}$ is a solution of the three-dimensional Laplace equation $u_{xx} + u_{yy} + u_{zz} = 0$.
6. Let $w = x^2z + y^2z^3$ and $y = \sin 2x$ & $z = e^{2xy}$, Find $\frac{dw}{dx}$.
7. Verify that the function $u(x,t) = e^{-\alpha^2k^2t}\sin kx$ is a solution of the heat conduction equation $u_t = \alpha^2u_{xx}$.
8. $w = f(u^2 - v^2, u^2 + v^2, \tan^{-1}(u))$, apply chain rule to Find $\frac{\partial w}{\partial v}$ & $\frac{\partial w}{\partial u}$.

Maxima and Minima of functions of two variables

Determine all critical point(s). Also find the local maximum and minimum values and saddle points (if any) of the given function,

1. $f(x,y) = xy - \frac{1}{2}y^2 - \frac{1}{4}x^4$.
2. $f(x,y) = 2x^3 - 6x + 6xy^2$.
3. $f(x,y) = 4xy + 2y^4 + 2x^2$.

4. $f(x,y) = x^3y - 2y^3 + 3xy^2$.

Taylor expansion of multi-variable functions

1. Expand $f(x,y) = x^2y + 3y - 2$ in power of (x-1) and (y+2).
2. Expand $f(x,y) = e^x \ln(1+y)$ in power of x and y.
3. Expand $f(x,y) = \sin xy$ about $\left(1, \frac{\pi}{2}\right)$.

Best of Luck.